

BHABHA UNIVERSITY, BHOPAL

BACHELOR'S OF COMPUTER APPLICATIONS (BCA)



SYLLABUS BCA (EFFECTIVE FROM JULY 2022)

Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL
NH-12, Hoshangabad Road, Jatkhedi, Bhopal-462026

BHABHA UNIVERSITY, BHOPAL

SCHEME OF EXAMINATION BCA

Semester I

SNO	PAPER TYPE	NO. OF PAPERS	CREDIT	LECTURE HRS PER WEEK		MAX MARKS	
				THEORY	LAB	THEORY (UNIVERSITY EXAM +CCE)	LAB UNIVERSITY EXAM +CCE)
1	MAJOR (PROGRAMMING IN C) BCA11	1	6	3		100 (60+40)	
	PRORAMMING IN C (LAB) BCA16	LAB			4		50
2	MINOR (FUNDAMENTALS OF COMPUTER) BCA12	1	6	3		100 (60+40)	
	FUNDAMENTALS OF COMPUTER LAB BCA17	LAB			4		50
3	GEC(GENERIC ELECTIVE COURSE) COMPUTATIONAL MATHEMATICS BCA13	1	4	3		100 (60+40)	
4	AEC(ABILITY ENHANCEMENT COURSE) (ENGLISH) BCA14	1	4	3		50 (30+20)	
	AEC(ABILITY ENHANCEMENT COURSE) (HINDI) BCA15	1		3		50 (30+20)	
5	ADL OFFICE TOOLS A1	1	4	3		100 (60+40)	
	TOTAL	6	24			500	100
	Grand Total						600

Semester II

SNO	PAPER TYPE	NO. OF PAPERS	CREDIT	LECTURE HRS PER WEEK		MAX MARKS	
				THEORY	LAB	THEORY (UNIVERSITY EXAM +CCE)	LAB UNIVERSITY EXAM +CCE)
1	MAJOR (DATA STRUCTURE WITH C++) BCA21	1	6	3		100 (60+40)	
	DATA STRUCTURE WITH C++ LAB BCA25	LAB			4		50
2	MINOR (DIGITAL COMPUTER ORGANIZATION) BCA22	1	6	3		100 (60+40)	
	DIGITAL COMPUTER ORGANIZATION LAB BCA26	LAB			4		50
3	GEC(GENERIC ELECTIVE COURSE) PROBABILITY AND STATICS BCA23	1	4	3		100 (60+40)	
4	AEC(ABILITY ENHANCEMENT COURSE) ENVITRONAMENTAL EDUCATION) BCA24	1	4	3		100 (60+40)	
5	ADL DTP A2	1	4	3		100 (60+40)	
	TOTAL	5	24			500	100
	Grand Total						600

Semester III

SNO	PAPER TYPE	NO. OF PAPERS	CREDIT	LECTURE HRS PER WEEK		MAX MARKS	
				THEORY	LAB	THEORY (UNIVERSITY EXAM +CCE)	LAB UNIVERSITY EXAM +CCE)
1	MAJOR (DBMS) BCA31	1	6	3		100 (60+40)	
	DBMS LAB BCA35	LAB			4		50
2	MINOR (OPERATING SYSTEM) BCA32	1	6	3		100 (60+40)	
	OPERATING SYSTEM LAB BCA36	LAB			4		50
3	GEC(GENERIC ELECTIVE COURSE) ORGANIZATION BEHAVIOUR BCA33	1	4	3		100 (60+40)	
4	SEC(SKILL ENHANCEMENT COURSE) PHP & MY SQL BCA34	1	4	3		100 (60+40)	
5	ADL DIGITAL MARKETING A3	1	4	3		100 (60+40)	
	TOTAL	5	24			500	100
	Grand Total						600

Semester IV

SNO	PAPER TYPE	NO. OF PAPERS	CREDIT	LECTURE HRS PER WEEK		MAX MARKS	
				THEORY	LAB	THEORY (UNIVERSITY EXAM +CCE)	LAB UNIVERSITY EXAM +CCE)
1	MAJOR (JAVA PROGRAMMING) BCA41	1	6	3		100 (60+40)	
	JAVA PROGRAMMING LAB BCA45	LAB			4		50
2	MINOR (SYSTEM ANALYSIS AND DESIGN) BCA42	1	6	3		100 (60+40)	
	SYSTEM ANALYSIS AND DESIGN LAB BCA46	LAB			4		50
3	GEC(GENERIC ELECTIVE COURSE) NUMERIC ANALYSIS BCA43	1	4	3		100 (60+40)	
4	SEC(SKILL ENHANCEMENT COURSE) FINANTIAL ACCOUTING & MANAGEMEN BCA44	1	4	3		100 (60+40)	
5	ADL MANEGERIAL ECONOMICS A4	1	4	3		100 (60+40)	
	TOTAL	5	24			500	100
	Grand Total						600

Semester V

SNO	PAPER TYPE	NO. OF PAPERS	CREDIT	LECTURE HRS PER WEEK		MAX MARKS	
				THEORY	LAB	THEORY (UNIVERSITY EXAM +CCE)	LAB UNIVERSITY EXAM +CCE)
1	MAJOR (COMMUNICATION & NETWORK) BCA51	1	6	3		100 (60+40)	
	COMMUNICATION & NETWORK LAB BCA55	LAB			4		50
2	FIELD PROJECT/ INTERNSHIP /APPRENTICESHIP/COMMUNITY SERVICE BCA52	LAB	6		8		150
3	SEC(SKILL ENHANCEMENT COURSE) INFORMATION STORAGE AND MANAGEMENT) BCA53	1	4	3		100 (60+40)	
4	DSE(DISCIPLINE SPECIFIC ELECTIVE) SOFTWARE ENGINEERING BCA54	1	4	3		100(60+40)	
5	ADL ARTIFICIAL INTELIGENCE A5	1	4	3		100 (60+40)	
	TOTAL	4	24			400	200
	Grand Total						600

Semester VI

SNO	PAPER TYPE	NO. OF PAPERS	CREDIT	LECTURE HRS PER WEEK		MAX MARKS	
				THEORY	LAB	THEORY (UNIVERSITY EXAM +CCE)	LAB UNIVERSITY EXAM +CCE)
1	MAJOR (WEB DESIGNING) BCA61	1	6	3		100 (60+40)	
	WEB DESIGNING LAB BCA65	LAB			4		50
2	FIELD PROJECT/ INTERNSHIP /APPRENTICESHIP/COMMUNITY SERVICE BCA62	LAB	6				150
3	DSE(DISCIPLINE SPECIFIC ELECTIVE) (CLOUD COMPUTING) BCA63	1	8	3		100 (60+40)	
	DSE(DISCIPLINE SPECIFIC ELECTIVE) (THEORY OF COMPUTATION) BCA64	1				100 (60+40)	
4	ADL SOFTWARE TESTING AND QUALITY ASSURANCE A6	1	4	3		100 (60+40)	
	TOTAL	4	24			400	200
	Grand Total						600

BHABHA UNIVERSITY, BHOPAL

BACHELORS OF COMPUTER APPLICATIONS (BCA)



SYLLABUS BCA I SEMESTER (EFFECTIVE FROM JULY 2022)

Department of Computer Science & Application
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BHABHA UNIVERSITY, BHOPAL

SCHEME OF EXAMINATION BCA

Semester I

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	FUNDAMENTALS OF COMPUTER LAB BCA17	LAB			4		50
3	GEC(GENERIC ELECTIVE COURSE) COMPUTATIONAL MATHEMATICS BCA13	1	4	3		100 (60+40)	
4	AEC(ABILITY ENHANCEMENT COURSE) (ENGLISH) BCA14	1	4	3		50 (30+20)	
	AEC(ABILITY ENHANCEMENT COURSE) (HINDI) BCA15	1		3		50 (30+20)	
5	ADL OFFICE TOOLS A1	1	4	3		100 (60+40)	
	TOTAL	6	24			500	100
	Grand Total						600

BACHELORS OF COMPUTER APPLICATIONS

BCA11

Programming in C

Unit I	Introduction, Data Types and operators identifiers and keywords, constants, types of operators, type conversion, writing a C-Program, variable declaration, C-Statements, Input and Output functions	
Unit II	Control statement, conditional expressions if statement, if-else statement, case and switch statement, loop-statements: For loop, while loop, do while loop, Break, continue and go to statements	
Unit III	Functions and program structure, Function definition, Type of functions, local and global variables, scope of variables, multifunction programs, Recursive functions.	
Unit IV	Arrays Notation and declaration, initialization, multidimensional and character arrays, pointers, Declarations, Pointer arithmetic, pointers and functions.	
Unit V	Preprocessors and macros, Header files (brief introductions only), structures, Declarations, initialization and use of structures in a C-Program function and structures, Array of structures Arrays within a structures. Unions	

. TEXT & REFERENCE BOOKS:

- 1• HERBERT SHIELD, "COMPLETE REFERENCE C"
- 2• Y KANETKAR, "POINTERS THROUGH C".
- 3• Y KANETKAR, "TSR THROUGH C".
- 4• R.S SALARIA, "APPLICA TION PROGRAMMING IN C"

BACHELORS OF COMPUTER APPLICATIONS

BCA12

Fundamentals of Computer

Unit I	Introduction to computer system, block diagram of a computer, types of software's. Analog , digital & hybrid, general and special purpose computers. Computer generations.	
Unit II	Introduction to Input devices, output devices, Storage devices , data storage and retrieval methods-sequential, direct & indexed & sequential.	
Unit III	Data processing system, storing data, processing data. central processing unit the microprocessor control unit, ALU, Register, Buses , RAM, ROM . CRT and flat screen technologies.	
Unit IV	Introduction types of operating system programs, booting loader, diagnostic tests, operating system executive, bios, utility programs, file maintenance, language processors, assembler, compiler and interpreter, applications software: microcomputer software, interacting with system, trends in pc software, difference between program and packages.	
Unit V	computer programming languages, types of programming languages, generations of programming languages, low level , high level language.	
1. Text Books: COMPUTERS TODAY, BY S.K BASANDRA, GALGOTIA PUBLICATIONS. 2. FUNDAMENTALS OF INFORMATION TECHNOLOGY ALEXIS LEON & MATHEWS LEON.		

BACHELORS OF COMPUTER APPLICATIONS

BCA13

Computational Mathematics

Unit I	Trigonometry: Angles & their Measurements, values of trigonometric Ratios, Height and Distances. Elementary Matrices and types of Matrices.	
Unit II	Equations: Simultaneous linear equations, Methods of Solving Simultaneous equations, Quadratic equations.	
Unit III	Statistics: Frequency Distribution, Measures of Central Tendency: Mean, Mode, Median Measures of Variations: Mean Deviation Standard Deviation.	
Unit IV	Mathematical Logic: Statements and notations, Connectives: Negation, Conjunctions and Disjunction. Statements formulas and truth tables . Tautologies, Tautological implications, contradiction contingency.	
Unit V	Set Theory: Basic concepts of set theory, notation, inclusion and equality of sets, the power set, types of sets, operations on set, Venn diagrams.	

Text Books:

1. Business Mathematics S.M. Shukla, Sahitya Bhawan Publications.
2. Business Mathematics D.C.Agrawal, Sree Sai Prakashan.
3. S.K.Sarkar : A Text Book of Discrete Mathematics S.Chand, 2005
4. A text book of Discrete Mathematics 9/E, Sarkar S.K, S. Chand New Delhi, 2016
5. Discrete Mathematics Structures with application to computer science”. Indian Edition, J.P Tremblay, R.Manohar, McGraw Hill Education 2017
6. “Discrete Mathematics”, 2/E, J.K.Sharma, Macmillan Publication, 2005

BACHELORS OF COMPUTER APPLICATIONS

BCA14

English

	Topics	
UNIT-I	Reading writing and interpretation skill 1 Where The Mind Is Without Fear - Rabindranath Tagore [keyword :patriotism] 2 National Education - M. K. Gandhi [keyword : Edification] 3 The Axe - R K Narayan [keyword : Environment] 4 The Wonder That was India- A. L. Basham (an excerpt)[keyword: Indianess] 5 Preface to Mahabharata C Rajagopalachari [keyword: Indian mythology]	
UNIT-II	comprehension skill unseen passage followed by multiple choice question	
UNIT-III	Basic language skill 1 vocabulary building :suffix, prefix ,synonym, antonyms and one word substitution 2 Basic grammar : noun ,pronoun, adjective verb, adverb ,preposition, article	
TEXT & REFERENCE BOOKS: suggested readings Essential English grammar - Raymond Murphy , Cambridge University Press Practical English grammar exercises 1 A J Thomson and AV Martinate, Oxford India practical English usage - Michael Swan, Oxford English grammar and use - Raymond Murphy , Cambridge University Press		

BACHELORS OF COMPUTER APPLICATIONS

BCA15

हिंदी

इकाई एक	मैथिलीशरण गुप्त -जीवन परिचय , मातृभूमि (कविता) प्रेमचंद- जीवन परिचय , शतरंज के खिलाड़ी (कहानी) शरद जोशी -जीवन परिचय , जीप पर सवार इल्लियां (व्यंग)	
इकाई दो	वैचारिक - भारतीय भाषाओं में राम आचार्य रामचंद्र शुक्ल- जीवन परिचय , पाठ उत्साह (भाव मूलक निबंध) रामधारी सिंह दिनकर जीवन परिचय, पाठ भारत एक है (संस्कृति) आदि शंकराचार्य - जीवन व दर्शन	
इकाई तीन	व्याकरण: पर्यायवाची शब्द, विलोम शब्द , अनेक शब्द के लिए एक शब्द , संधि और उसके प्रकार भाषा विकास , भाषा परिभाषा अवधारणा का अर्थ एवं परिभाषा	
पाठ्य पुस्तक , संदर्भ पुस्तक , अन्य संसाधन है प्रेमचंद मानसरोवर खंड 3 आचार्य रामचंद्र शुक्ल चिंतामणि भाग 1 डॉ वासुदेव नंदन प्रसाद आधुनिक हिंदी व्याकरण और रचना भारती भवन ठाकुरबाड़ी रोड पटना बिहार डॉक्टर राजेश्वर चतुर्वेदी हिंदी व्याकरण उपकार प्रकाशन आगरा उत्तर प्रदेश हिंदी ज्ञानकोष इंटरनेट सामग्री		

BACHELORS OF COMPUTER APPLICATIONS

A1 Office Tools

Unit I	MS Word :Introduction, feature and area of use., working with MS Word , Ribbon Tabs : Home, insert, page layout references, mailing, review and view etc.. using word to create a new document customization of ribbon ,open ,save and print a document, edit and format text ,change the page layout background and borders, insert and edit tables. insert Clipart and pictures to document. formatting fonts in word . applying text effects using character spacing border and colors . inserting header and footer . using date and time option in word . creating project abstract features to be covered :formatting style, inserting table bullets and numbering ,changing text direction . cell alignment, footnotes ,hyperlink symbols, spell check ,track changes	
Unit II	Creating a news letter : features to be covered - table of content , newspaper columns image form files and Clipart, drawing toolbars and word art, formatting images text boxes and paragraphs ,creating a feedback form. features to be covered forms ,text fields inserting objects. mail merge creating custom document ,creating main document, creating data source .editing data source, opening a data source. sorting the data source ,finding a record of data source, editing main document ,sorting merge documents ,filtering Merge documents. printing merged documents, merging want to letterhead using different data sources with a single main document.	
Unit III	MS Excel :introduction to excel interface ,understanding rules and columns ,naming cells ,working with Excel workbook and worksheet formatting Excel workbook, new ,open, close ,save, save as ,formatting text: font size ,font style, font colour ,use the bold ,Italic and underline .wrap text, merge and centre , currency account and other formats modifying columns rows and cells, perform calculations with functions ,creating simple formulas setting up your own formula date and time functions. financial functions logical functions look up and references. Functions calculations features to be covered -cell referencing formula in Excel -average, standard deviation, charts ,renaming and inserting worksheets . hyper linking. count functions. mathematical functions statistical functions. text functions. sort and filter data with Excel ,sort and filtering data using number filter ,text filter, custom filtering, removing filters from columns, conditional formatting	
Unit IV	Create effective charts to present data visually, inserting columns, pie chart etc . create an effective chart with chart tool ,design, format and layout options ,adding chart title, changing layout, chart style, editing chart data range ,editing data series, trendlines ,protecting and sharing the workbook ,protecting a workbook with a password . allow user to edit ranges ,track changes. working with comments. performance analysis -features to be covered : split cells, freeze panes ,group and outline. sorting, boolean and logical operators ,conditional formatting . cricket score card creations -features to be covered pivot tables, interactive buttons ,importing data, data protection, data validation ,what if analysis	

Unit V	Creating PowerPoint presentations : making presentation which demonstrate use of hyperlinks, inserting images ,Clipart, audio, video, objects, table and charts. create master layouts (slide, template and notes, types of views(basic, presentation ,slide sorter, notes etc). inserting -background textures ,design templates, hidden slides ,auto content Wizard ,slide transition, custom animation, auto rehearsing	

Text Books & Reference Books:

:

1. Taxali. "PC Software", IMIT
2. Saxena Sanjay. S Sehind, "Office Tools 2000 for 1 veryone".
3. Kari Holloway. "Write's Guide to Microsoft Word",
4. Michael Alexander, Richard Kamleika. Grey Harvey, Excel 2019
5. . Chris Smith, "Microsoft PowerPoint Made Easy"
6. . Sini Shibu, Shilpa Jackson, "Computer Basics, MS-Office & Internet", BSSS Publications

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SYLLABUS BCA II SEMESTER (EFFECTIVE FROM JULY 2022)

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BHABHA UNIVERSITY, BHOPAL

SCHEME OF EXAMINATION BCA

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BACHELORS OF COMPUTER APPLICATIONS

BCA21

DATA STRUCTURE WITH C++

Unit I	Definition of Data Structure, its types, Basic operations in Data Structures. Array – its representation, Operations on arrays, address calculation of single and multi dimensional arrays in memory, Applications of arrays.	
Unit II	Stack: Representation of Stacks, operations on Stacks, infix and postfix notations, postfix expression evaluation, application of Stacks. Queues: Representation of queues, operations on queues, circular queue, Dequeue, Priority Queue, Applications of queues.	
Unit III	Linked list: Single Linked List – description & operations, Doubly Linked List – description & operations, Linked implementation of Stacks and queues, Header linked list.	
Unit IV	Trees: Basic terminology, binary tree, binary search tree, Operations on binary tree: Insertion & Deletion algorithms. Traversal of binary trees: Inorder, Preorder & Post order.	
Unit V	Searching: Linear Search, Binary Search. Sorting: Selection Sort, Bubble Sort method & its algorithm, Quick Sort, Merge Sort. Graphs: Related Concepts and its representations. Graph Traversal Schemes: Depth first search, Breadth first search.	
. TEXT BOOKS: 1. Data Structure by Schaum Series. 2. Data Structure by Tanenbaum. 3.Data Structure by Sartaj Sahani 4. Data Structure using C++ by Yashwant Kanetkar		

BACHELORS OF COMPUTER APPLICATIONS

BCA22

DIGITAL COMPUTER ORGANIZATION

Unit I	Data Representation Data Types and Number Systems, Binary Number System, Octal & Hexa-Decimal Number System, Fixed Point Representation, 1's & 2's Complement, Binary, Arithmetic Operation on Binary Numbers, Overflow & Underflow, Floating Point Representation, Codes, ASCII, EBCDIC Codes, Gray Code, Excess-3 & BCD, Error Detection & Correcting Codes Binary Storage and Registers.	
Unit II	Boolean algebra and Digital logic circuits -Logic Gates, AND, OR, NOT, NOR, NAND & XOR Gates and their Truth Tables, Boolean Algebra, Basic Definition and Properties, Basic Boolean Law's, Demorgan's Theorem, Minimization Techniques, K Map – Two, Three and More Variables maps, Sum of Product & Product of Sums, Don't care conditions.	
Unit III	Combination Circuits - Half adder & Full adder, Full Subtractor, Full Subtractor and decimal adder, Code Conversion, Multilevel NAND and NOR Circuits, Decimal adder, decoders, Multiplexers and Demultiplexers.	
Unit IV	Sequential logic- Flip-Flops - RS, D, JK & T Flip-Flop, Triggering in flip flops, Analysis of Clocked Sequential Circuits, State Reduction and Assignment, flip flop excitation tables, Design procedure and design of counters. Design with equations.	
Unit V	Registers, Counters and the memory unit, Shift registers, Ripple counters and Synchronous counters, Inter-register Transfer, Arithmetic Logic and Shift Micro Operation, Conditional Control Statement, Instruction Codes, Processor organization, design of a simple computer.	

TEXT & REFERENCE BOOKS:

1. DIGITAL LOGIC AND COMPUTER DESIGN BY MORRIS MANO
2. COMPUTER SYSTEM ARCHITECTURE BY MORRIS MANO

BACHELORS OF COMPUTER APPLICATIONS

BCA23

PROBABILITY AND STATICS

Unit I	Theory of Probability - I: Event and Sample space, Probability of an event, Addition multiplication theorem of probability, Inverse probability, Bay's theorem. Continuous probability.	
Unit II	Theory of Probability - II: Probability density function and its applications, Standard deviation of various continuous probability distributions, Mathematical expectation, Expectation of sum and product of random variables.	
Unit III	Measures of dispersion: Range and inter quartile range, Mean deviation and Standard deviation, Moments, Skewness and kurtosis. Moment generating function. Theoretical distribution: Binomial, Poisson, Rectangular, Exponential.	
Unit IV	Curve fitting and Correlation: Methods of least squares, Curve fitting, Correlation and regression, Partial and multiple correlations (up to three variables only)	
Unit V	Sampling: Sampling of large samples, Null and alternative hypothesis, Errors of first and second kinds, Level of significance and critical region, Tests of significance based on chi-square (χ^2), t, F and Z distribution.	

Text Books. Reference Books

- 1, H. C. Saxena and J. N. Kapoor: Mathematical Statistics, S. Chand and Company, 2010.
2. E. Rukmangadachari: Probability and Statistics, Pearson Education India; First edition, 2012.
3. Vrajay I(. Rohatgi, A. K. Md. EhsanesSaleh: An Introduction to Probability and Statistics, Wiley; 3rd edition, 2015.
4. S. C" Gupta and V. K. Kapoor: Fundamentals of Mathematical Statistics, Sultan Chand & Sons, 2014.

BACHELORS OF COMPUTER APPLICATIONS

BCA24

ENVIRONMENTAL EDUCATION

	Topics	
UNIT-I	Environment and Natural Resources: Multidisciplinary nature, Scope and Importance of Environment. Components of Environment: Atmosphere, Hydrosphere, Lithosphere, and Biosphere. Brief account of Natural Resources and associated problems: Land Resource. Water Resource. Energy Resource. Concept of Sustainability and Sustainable Development	
UNIT-II	Biome, Ecosystem and Biodiversity: Major Biomes: Tropical, Temperate. Forest, Grassland, Desert. Tundra, Wetland. Estuarine and Marine. Ecosystem: Structure function and types their Preservation & Restoration Biodiversity and its conservation practices.	
UNIT-III	Environmental Pollution: Pollution: Types, Control measures, Management and associated problems. • Environmental Law and Legislation: Protection and conservation Acts. • International Agreement & Programme. A brief account of air, water, soil and noise pollutions; Biological, industrial and solid wastes in urban areas. Human health and economic risks.	
Unit IV	Management and Social Issues: Environmental Movements, communication and public awareness programme. National and International organizations related to environment conservation and monitoring. Role of information technology in environment and human health. Non-renewable energy resources, their utilization and influences. Green house effect - global warming; ocean acidification, ozone layer depletion, acid rains and impacts on human communities and agriculture.	
Unit V	Environmental degradation and impacts : Human population growth and its impacts on environment; land use change, land degradation, soil erosion and desertification. Use and over-exploitation of surface and ground water, construction of dams, floods, conflicts over water (within India). Deforestation: Causes and effects due to expansion of agriculture, firewood, mining, forest fires and building of new habitats.	
	Suggested activities to learner: (4 hours) 1. Visit to an area to document environmental assets: river/ forest/ flora/fauna, etc 2. Visit to a local polluted site-Urban/Rural/Industrial/Agricultural site. 3. Study of common plants, insects, birds and basic principles of identification. 4. Study of simple ecosystems-forest, tank, pond, lake ,mangroves etc.	

5. Case study of a Forest ecosystem or a pond ecosystem.

Text book & Reference books: ErachBarucha (2004) Text book of Environmental Studies for Undergraduate courses→ (Prepared for University Grants Commmission) Universities Press.
PurnimaSmarath (2018) Environmental studies Kalyani Publishers, Ludhiana.
Singh, J.S., Singh, S.P. and Gupta, S.R. (2014) Ecology, Environmental Science and→ Conservation. S. Chand Publishing, New Delhi.
Sengupta, R. (2003) Ecology and economics: An approach to sustainable development.→ OUP.
Wilson, E. O. (2006) The Creation: An appeal to save life on earth. New York: Norton.→

BACHELORS OF COMPUTER APPLICATIONS

A2 DESK TOP PUBLISHING

	Topics	
UNIT-I	DESK TOP PUBLISHING: Why Graphics, Various Types of Graphics Programs. Drafting Packages, Introductions to Printing, DTP Package, Advantage of D.T.P in Publication, Microsoft Windows. Various Documentation cum DTP Package e.g. Microsoft Word Etc.	
UNIT-II	Introduction To PageMaker / Ventura or a similar Package. Page Maker Icon and help, Tool Box, Styles, Menus etc., Different screen Views, Importing text/Pictures Preparation of Document using DTP Package.	
UNIT-III	Text Formatting, Different Page Layouts, Printing Various Fonts And Characters Sets. Various Type of Printers used in DTP.	
Unit IV	Introduction to Commercial DTP System available in Market, Indian Language Fonts, Creation of Indian Language Fonts.	
Unit V	Import & Export of Documents created by other Word Processors, Spelling Check, Designing exercise like Visiting Card, Greeting Cards etc.	
	Reference Books Desktop Publishing on PC by MC Shrama, BPB Publication. PA GE MAKER 4.0 & 5.0 BY B.P.O. PUBLICATIONS 1. PAGEMAKER – MANUAL 2. „O“ LEVEL MODULE M3.2 DESKTOP PUBLISHING & PRESENTATION GRAPHICS BY V K JAIN, BPB PUBLICATIONS.	

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SYLLABUS BCA III SEMESTER (EFFECTIVE FROM JULY 2022)

Department of Computer Science & Applications
BHABHA UNIVERSITY, BHOPAL
NH-12, Hoshangabad Road, Jatkhedhi, Bhopal-462026

BHABHA UNIVERSITY, BHOPAL

SCHEME OF EXAMINATION BCA

Semester III

SNO	PAPER TYPE	NO. OF PAPERS	CREDIT	LECTURE HRS PER WEEK		MAX MARKS	
				THEORY	LAB	THEORY (UNIVERSITY EXAM +CCE)	LAB UNIVERSITY EXAM +CCE)
1	MAJOR (DBMS) BCA31	1	6	3		100 (60+40)	
	DBMS LAB BCA35	LAB			4		50
2	MINOR (OPERATING SYSTEM) BCA32	1	6	3		100 (60+40)	
	OPERATING SYSTEM LAB BCA36	LAB			4		50
3	GEC(GENERIC ELECTIVE COURSE) ORGANIZATION BEHAVIOUR BCA33	1	4	3		100 (60+40)	
4	SEC(SKILL ENHANCEMENT COURSE) PHP & MY SQL BCA34	1	4	3		100 (60+40)	
5	ADL DIGITAL MARKETING A3	1	4	3		100 (60+40)	
	TOTAL	5	24			500	100
	Grand Total						600

BACHELOR'S OF COMPUTER APPLICATIONS

BCA31

Database Management System

Unit I	Concepts of DBMS: Data, Information, Database, Components of DBMS, Architecture of a database system – Physical, Conceptual and User level, Data Independence – Logical and Physical, DBMS terminology, Data Dictionary. Database Models: Network, Hierarchical and Relational Models, Brief idea about the new concepts like distributed databases, parallel databases.	
Unit II	RDBMS: Definition, Components, Terminology, Difference with DBMS. ER-Model, ER-Diagram, ER-concepts, types of relationships. Normalization: Introduction, concepts associated with normalization. key terminology: primary, candidate, foreign, alternate keys.	
Unit III	Idea about Generalization, Aggregation, Specialization. Relational Algebra: Formal Definition, Fundamental Operations – select, project, union, set, difference, Concept of SQL sublanguages – DDL, DML, DCL, TCL, SCL etc., DML commands: Insertion, updation, deletion operations, many faces of select command, data constraints, logical operators	
Unit IV	Functional Dependencies: basic definitions, trivial and non trivial dependencies, closure set of dependencies and of attributes, irreducible set of dependencies, , non loss decomposition, FD diagram, dependency preservation, Transaction, concurrency and Recovery: basic concepts, ACID properties, Transaction states, implementation of atomicity and durability, concurrent executions, basic idea of serializability, basic idea of concurrency control, basic idea of deadlock, failure classification.	
Unit V	Storage structure and file organizations: overview of physical storage media, magnetic disk performance and optimization, basic idea of RAID, file organization, organization of records in files, basic concepts of indexing, ordered indices, basic idea of B-tree and B+-tree organization Network and hierarchical models: basic idea, data structure diagrams, DBTG model, implementations, tree structure diagram, implementation techniques, comparison of the three models.	

TEXT & REFERENCE BOOKS:

1. Database System Concepts – by Silberschatz, Korth, Sudarshan (Mc Graw Hill).
2. An Introduction to Database Systems – by Bipin Desai.
3. SQL, PL/SQL – The Programming language of Oracle – by Ivan Bayross (BPB).
4. Introduction to Database Systems by C J Date (Pearsons Education).
5. SQL/PLSQL for Oracle by P.S.Deshpande (AWI).

BACHELOR'S OF COMPUTER APPLICATIONS

BCA32 OPERATING SYSTEM

Unit I	Operating System Definitions, its Components, Evolution of Operating System, types of operating systems: batch, multi programmed, multitasking, desktop, multiprocessor, real-time, client-server, peer-to-peer, distributed, clustered and handheld.
Unit II	Process Scheduling: concept of a process, process states, PCB, Process state transitions, operations on processes, context switch, types of schedulers, CPU scheduling concept, CPU Scheduler, CPU-I/O burst cycle, dispatcher, scheduling criteria, scheduling algorithms – FCFS, SJF, STRN, Round Robin, priority, multilevel queue and event driven (i.e., pre-emptive priority) scheduling algorithms.
Unit III	Memory Management Concepts: Address Binding, logical and physical address space, dynamic loading etc., Contiguous allocation methods – static & dynamic partitioned memory allocation. Concepts of fragmentation, swapping, relocation, compaction, protection, sharing. Segmentation. Non-contiguous allocation methods – Paging: basic principle of operation, h/w support for paging, protection and sharing. Virtual memory: concept of demand paging, Page fault, page replacement algorithms – FIFO, LRU, OPT.
Unit IV	File system implementation, Responsibilities of file management system, directory implementation as linear list/hash table, directory structure, disk organization, disk controller and driver, disk space management – contiguous allocation, non contiguous allocation – chaining and indexing, disk address translation.
Unit V	Device Management: I/O hardware, Techniques for device management. Dedicated devices, shared devices, virtual peripherals. Security & protection: Security threats and goals, penetration attempts, Security policies and mechanism, authentication, protection and access control. Interprocess communication, need for inter process synchronization. Deadlocks – definition, avoidance, detection, prevention and recovery.

TEXT & REFERENCE BOOKS:

1. Operating System Concepts – by Silberschatz, Galvin and Gagne.
2. Operating System Concepts and Design – by Milenkovic
- 3.. Operating System – by Tanenbaum.
4. Operating System – by Peterson

BACHELOR'S OF COMPUTER APPLICATIONS

BCA33

ORGANIZATION BEHAVIOUR

Unit I	Organization , What is Psychology, Whom does the Psychologist observe, What does Psychologist observe, Where does the Psychologist observe,	
Unit II	Sensation & Perception, Feelings and Emotions, Learning and Thinking.(in so far as they are applicable to Industry), Personality, Definition & Theories, Assessing Personality.	
Unit III	An Overview of Transactional Analysis as a Tool for Measuring One's Own Awareness and of helping Interpersonal Relationship in an Organization. Theories of Organization, Contingency Model and other Models of Organization, Theories of Motivation. – Abraham Maslow, Herzerberg Mecleland.	
Unit IV	Group Dynamics, Structure, Process, Values of Groups, Role and Stature. Theories of Leadership, Identifying Leadership Potential.	
Unit V	Communication: Process, Methods, Barriers and Usefulness.	
TEXT & REFERENCE BOOKS: 1. Organizational Behavior – by Robbins. 2. Organizational Behavior – by Luthans. 3. Organizational Behavior – by Sashi Gupta & Rozy Joshi.		

BACHELOR'S OF COMPUTER APPLICATIONS

BCA34
PHP& MY SQL

	Topics	
UNIT-I	Basic of PHP :Introduction to PHP what does PHP do ,object oriented programming with PHP, language basics, installation of XAMP/LAMP , syntax, comments, variables ,constant and data types ,expression and operators, flow control statement including HTML code in PHP ,embedding PHP in web pages.	
UNIT-II	Functions and strings :Definition of function, calling a functions variable scope, function parameters , return values, predefined functions. string: creating and accessing string ,searching and replacing string ,encoding and escaping ,comparing string, formatting string, regular expression.	
UNIT-III	Data and file handling :PHP Forms, \$Get ,\$ POST ,\$ REQUEST, \$ FILES,\$= SERVER, \$ GLOBALS ,\$ ENV input/ output controls, validations, cookies and sessions. File handling: file and directory open, close, read, write, append, delete, uploading and downloading files. file exist, file Si rename, reading and display all/ selected files present in a directory.	
Unit IV	My SQL and overview :Introduction, Php MYadmin, entering queries ,creating and using database, creating and selecting database, creating a table , loading data into a table, retrieving information from a Table , selecting all data, selecting particular row , selecting particular columns , sorting date, columns working with null values ,pattern machine ,counting row, using more than one table	
Unit V	My SQL DATABASE IN PHP: Introduction connecting to a MY SQL database, querying in database, retrieving and displaying the result ,modifying data and deleting data through front end. ,designing application using PHP& my SQL.	
TEXT & REFERENCE BOOKS: PHP& MY SQL , Vikram Vaswani, McGraw-Hill The complete reference - PHP , Steven Holzner, McGraw-Hill The complete reference - MY SQL , Vikram Vaswani, McGraw-Hill		

BACHELOR'S OF COMPUTER APPLICATIONS

A3

DIGITAL MARKETING

	Topics	
UNIT-I	Introduction to digital marketing ,meaning of digital marketing, evaluation of digital marketing from tradition to Modern era, traditional marketing versus Digital Marketing ,concept, scope and importance of digital marketing. Increasing visibility , visitor's engagement ,brining target targeted traffic, lead generation conversion retention. Emergence of digital marketing as a tool ,digital marketing channel -types and business model, digital marketing applications and benefits .	
UNIT-II	Challenges And opportunities for digital marketing, digital penetration in the Indian market. Ethical issues and legal challenges in digital marketing , regulatory Framework for Digital Marketing in India, digital marketing landscape: an overview Internet basics :internet / intranet/ portals ,understanding wave and wireless Technology. concept and role of Internet in marketing, online marketing domains, website design and domain name branding.	
UNIT-III	Digital marketing presence: digital marketing mix , the POEM framework CMS ,design of website, word-press domain registration and hosting setting up and installation ,overview of dashboard ,page ,post categories, menu ,various widgets plugins ,themes, Yoast ,SEO ,RSS including multimedia like audio video and image into the website.	
Unit IV	Social media marketing, Face book for business, Face book fan engagement , anatomy of ad campaign, Facebook insight and other marketing tools, Facebook Analytics. YouTube: introduction to YouTube marketing, YouTube marketing strategy ,the subscriber advantage -account setup ,Optimization ,keyword research and video structure, YouTube SEO ,Annotations and Cards ,YouTube promotion and analytics, YouTube monetization	
Unit V	Twitter marketing: Twitter marketing getting started with twitter ,twitter marketing components , Twitter building blocks, building up content strategy. Tends marketing on twitter ,Twitter usage customer service, Twitter ads campaign types targeting audience options , Twitter ads pricing, metrics, Twitter Analytics, Introduction to LinkedIn: LinkedIn present builder robust company page on LinkedIn , sales leads generation using LinkedIn video marketing: tools and technique mobile marketing tools.	
TEXT & REFERENCE BOOKS: 1. Chaffey, D.. Chadwick, F. E., Johnston, K., & Mayer, R. (2008), Internet Marketing: Strategy, Implementation, and Practice. New Jersey. Pearson Hall. 2. Frost, R. D.. Fox, A. & Strauss, J. (2018). E-Marketing. Abingdon: Routledge. Gupta, S. (2018). Digital Marketing. Delhi: Tata McGraw Hill Education.		

3. Kapoor, N. (2018). Fundamentals of E-Marketing. Delhi: Pinnacle India.

4. Kotler, P., Kartajaya, H., & Setiawan, 1 (2017) Digital Marketing: 4.0 Moving from Traditional to Digital, New Jersey in Wiley & Sons.

BHABHA UNIVERSITY, BHOPAL

BACHELOR'S OF COMPUTER APPLICATIONS (BCA)



SYLLABUS BCA IV SEMESTER (EFFECTIVE FROM JULY 2022)

Department of Computer Science & Applications
BHABHA UNIVERSITY, BHOPAL
NH-12, Hoshangabad Road, Jatkhedi, Bhopal-462026

BHABHA UNIVERSITY, BHOPAL

SCHEME OF EXAMINATION BCA

Semester IV

SNO	PAPER TYPE	NO. OF PAPERS	CREDIT	LECTURE HRS PER WEEK		MAX MARKS	
				THEORY	LAB	THEORY (UNIVERSITY EXAM +CCE)	LAB UNIVERSITY EXAM +CCE)
1	MAJOR (JAVA PROGRAMMING) BCA41	1	6	3		100 (60+40)	
	JAVA PROGRAMMING LAB BCA45	LAB			4		50
2	MINOR (SYSTEM ANALYSIS AND DESIGN) BCA42	1	6	3		100 (60+40)	
	SYSTEM ANALYSIS AND DESIGN LAB BCA46	LAB			4		50
3	GEC(GENERIC ELECTIVE COURSE) NUMERIC ANALYSIS BCA43	1	4	3		100 (60+40)	
4	SEC(SKILL ENHANCEMENT COURSE) FINANTIAL ACCOUTING & MANAGEMEN BCA44	1	4	3		100 (60+40)	
5	ADL MANEGERIAL ECONOMICS A4	1	4	3		100 (60+40)	
	TOTAL	5	24			500	100
	Grand Total						600

BACHELOR'S OF COMPUTER APPLICATIONS

BCA 41
JAVA PROGRAMMING

Unit I	ooPs concepts – Data abstraction, encapsulation, inheritance, benefits of inheritance, polymorphism. The Java Environment: Setting Class path; Data types; Operators - precedence and associativity ; Type conversion; Control and Iterative statements; Modular programming methods.	
Unit II	Object Oriented Programming in Java: Class; Objects; Packages; Scope and lifetime; Access Modifiers; Constructors; Copy constructor; this pointer; finalize () method; Arrays; Memory allocation and garbage collection Inheritance : Inheritance basics, method overriding, dynamics method dispatch, abstract classes. Interfaces : Defining an interface, implementing & applying interfaces, variables in interfaces, extending interfaces.	
Unit III	Multithreading and Exception Handling: Basic idea of multithreaded programming; The lifecycle of a thread; Creating thread with the thread class and runnable interface; Thread synchronization; Thread scheduling; Producer-consumer relationship; Daemon thread, Selfish threads; The try, catch and throw; throws Constructor and finalizers in exception handling; Applets: Applet security restrictions; the class hierarchy for applets; Life cycle of applet; HTML Tags for applet.	
Unit IV	Input/ Output : Exploring Java I/O, Directories, stream classes The Byte stream : Input stream, output stream., JDBC: JDBC-ODBC bridge; The connectivity model; The driver manager; Navigating the result set object contents; java.sql Package; The JDBC exception classes; Connecting to Remote database. Collections: The collections framework, collection interfaces, collection classes	
Unit V	AWT Fundamentals: The class hierarchy of window fundamentals; The basic user interface components , Frame, Layout managers, flow layout etc. The Java Event Handling Model: Java's event delegation model , Event class hierarchy; Adapter classes; Event classes action and different Events	

TEXT & REFERENCE BOOKS:

1. Naughton & Schildt "The Complete Reference Java 2", Tata McGraw Hill
2. Deitel "Java- How to Program:" Pearson Education, Asia
3. Horstmann & Cornell "Core Java 2" (Vol I & II) , Sun Microsystems
4. Ivan Bayross "Java 2.0" : BPB publications 5. Ivor Horton's "Beginning Java 2, JDK 5 Ed., Wiley India. Note : Paper is to be set unit wise with internal choice.
5. Internet and World Wide Web – How to program by Dietel and Nieto PHI/Pearson Education Asia.
6. Jakarta Struts Cookbook, Bill Siggelkow, S P D O'Reilly for chap 8
- 7 An Introduction to web Design and Programming –Wang Thomson
- 8 Web Applications Technologies Concepts- Knuckles, John Wiley

BACHELOR'S OF COMPUTER APPLICATIONS

BCA42

SYSTEM ANALYSIS AND DESIGN

Unit I	System Concept: Definition, Characteristics, Elements of system, Physical and abstract. System open and closed system, man-made information systems, System Development Life Cycle: Various phases of system development, Considerations for system planning and control for system success. System Planning	
Unit II	Tools of structured Analysis: Logical and Physical models context, diagram, data dictionary ,data diagram, form driven methodology, IPO and HIPO charts, Gantt charts, system model, pseudo codes, Flow charts, system flow chart, run flow charts etc., decision tree, decision tables, data validation. Management	
Unit III	System testing & quality: System testing and quality assurance, steps in system implementation and software maintenance. System security: Data Security, Disaster/ recovery and ethics in system development, threat and risk analysis, System audit. System Analyst	
Unit IV	The Software Product and Software Process: Software Engineering - A Layered Technology, Software Process Models: Linear Sequential Model. Prototyping Model, RAD Model Evolutionary Software Process Models: Incremental Model, Spiral Model Component Assembly Model.	
Unit V	Principles and Methods: The Software Design Process: Design Principles, Design Concepts, Effective Modular Design, Design Heuristics, Design Documentation, Design Methods: Data Design, Architectural Design, Interface Design, Human Computer Interface Design, Procedural Design.	
TEXT & REFERENCE BOOKS: 1. Computers Today – by Suresh K Basandra. 2. Modern System Analysis & Design by A Hoffer, F George, S Valaciah, Low price Edn. Pearson Education. 3. Introduction to Systems Analysis and Design – by Lee.		

BACHELOR'S OF COMPUTER APPLICATIONS

BCA43
NUMERIC ANALYSIS

Unit I	Solutions of Algebraic, Transcendental & Simultaneous Algebraic Equations. Introduction to Algebraic & Transcendental Equations, Newton-Raphson method, Regula-Falsi method, Successive Bisection method, Secant method, Comparison of Regula-Falsi method with Bisection method, Gauss Elimination method, Gause-Seidal method, Gauss Jordan method, Concept of Pivoting.	
Unit II	Interpolation & Extrapolation: Newton's Interpolation formulae, Newton's forward difference & Backward difference formula, Lagrange's Interpolation formula, Newton's divided-difference interpolation, Sterling's and Bessel's Central difference formula.	
Unit III	Numerical Integration & Differentiation: Introduction, Newton's Cotes formulae, Trapezoidal rule, Simpson's rule, Weddle's rule.	
Unit IV	Solution of ordinary differential & Integral Methods: Euler's, Picard's and Taylor's series methods. Modified Euler method, Picard's method for successive approximation, method of successive differentiation, Runge-kutta second & forth orders methods.	
Unit V	Correlation & Regression: Karl Pearson's Coefficient of Correlation, Rank Coefficient of Correlation, Partial & Multiple Correlation, Lines of Regression, Coefficient of Regression.	
TEXT & REFERENCE BOOKS: 1. Method of Numerical Methods by Shastri. 2. Computer Based Numerical Algorithms by Krishnamurthy. 3. Computer Oriented Numerical Methods by Salvadori. 4. Numerical Methods by H C Agarwal. 5. Fundamentals of Mathematical Statistics by Gupta & Kapoor.		

BACHELOR'S OF COMPUTER APPLICATIONS

BCA 44

FINANTIAL ACCOUTING & MANAGEMENT

	Topics	
UNIT-I	Introduction and Purpose of Accounting and Uses of Accounting Information, Basic Accounting Concepts, Accounting Structure, Process of Accounting, Journal, Ledger & Trial Balance based on Double Entry Book – Keeping.	
UNIT-II	Valuation of Assets and Depreciation Methods: Straight Line Method, Diminishing Balance Method, Sinking Fund Method, Insurance Method and Aunty Method.	
UNIT-III	Practical System of Accounting: Cash Book, Sales & Purchase of Goods, Bills of Exchange, Bank Reconciliation Statement.	
Unit IV	Preparation of Financial Statements: Income Statement (Profit & Loss Account), Statement of financial Position (Balance Sheet) and Adjustment.	
Unit V	Analysis of Financial Statements – Financial Ratio.	

TEXT & REFERENCE BOOKS:

1. Management Accounting By Hingorani, Grewal.
2. Financial Management by Khan & Jain.
3. Management Accounting by Shashi K Gupta, S Chand Publication.
4. Financial Accounts by Dr S M Shukla, Sahitya Bhavan Publications

BACHELOR'S OF COMPUTER APPLICATIONS

A4

MANAGERIAL ECONOMICS

	Topics	
UNIT-I	Managerial Economics: Introduction of Managerial Economics- Factors Influencing Manager, Micro and Macroeconomics, Theory of the Cost ,Theory of the Firm ,Theory of Production Function, Production System.	
UNIT-II	Input output Analysis, Micro Economics Applied to plants and industrial undertakings, Productivity, Factors affecting Productivity, Increasing Productivity of Resources	
UNIT-III	Meaning, Nature, Need, Role, Importance, Evolution of Management Through Information system. Structure of Management Information System. Relatedness of MIS with management activities, Management functions and decision making, Information System in Business and Management.	
Unit IV	Development of MIS- Methodology and Tools/ Techniques for systematic designing, implementation, evaluation. modification of MIS, A study of major financial, production, Man power and marketing MIS and case Studies.	
Unit V	Advanced MIS - concept need and problems in achieving advanced MIS, Decision support System. Rationale of computer application, Decision support system (DSS).	

TEXT & REFERENCE BOOKS:

1. Managerial Economics: Joel Dean: Prentice Hall of India
2. Murdick, RG, Ross,J.E. & Claggett, J.R.: Information systems for Modern Management, PHI
3. Thomas, R. & Prince: Information systems for planning and control.
4. Aktas ; structured analysis and design of information system, PHI
5. Spargue and Watson : Decision Support System ,2nd Edn. prentice Hall international ggg

BHABHA UNIVERSITY, BHOPAL

BACHELOR'S OF COMPUTER APPLICATIONS (BCA)



SYLLABUS BCA V SEMESTER (EFFECTIVE FROM JULY 2022)

Department of Computer Science & Applications
BHABHA UNIVERSITY, BHOPAL
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BHABHA UNIVERSITY, BHOPAL

SCHEME OF EXAMINATION BCA

Semester V

SNO	PAPER TYPE	NO. OF PAPERS	CREDIT	LECTURE HRS PER WEEK		MAX MARKS	
				THEORY	LAB	THEORY (UNIVERSITY EXAM +CCE)	LAB UNIVERSITY EXAM +CCE)
1	MAJOR (COMMUNICATION & NETWORK) BCA51	1	6	3		100 (60+40)	
	COMMUNICATION & NETWORK LAB BCA55	LAB			4		50
2	FIELD PROJECT/ INTERNSHIP /APPRENTICESHIP/COMMUNITY SERVICE BCA52	LAB	6		8		150
3	SEC(SKILL ENHANCEMENT COURSE) INFORMATION STORAGE AND MANAGEMENT) BCA53	1	4	3		100 (60+40)	
4	DSE(DISCIPLINE SPECIFIC ELECTIVE) SOFTWARE ENGINEERING BCA54	1	4	3		100(60+40)	
5	ADL ARTIFICIAL INTELLIGENCE A5	1	4	3		100 (60+40)	
	TOTAL	4	24			400	200
	Grand Total						600

BACHELOR'S OF COMPUTER APPLICATIONS

BCA 51
COMMUNICATION & NETWORK

Unit I	Data communication and IT, Communication Mode- Simplex, Half Duplex, Full Duplex, Communication Channels - Twisted, Coaxial, Fiber Optic, Serial and Parallel Communication, Types of Network - LAN, WAN, MAN ,Internet etc.	
Unit II	Topologies of LAN - Ring, Bus, Star, Mesh and Tree topologies, World Wide Web Internet Services, Analog & Digital Signal. Base Band , Broad Band, Multiplexer FDM, TDM, Modulation AM, FM, PM, Transmission Media ,Modem. OSI Reference Model , Switching Technique, Message Switching, Circuit Switching, Packet Switching, Virtual Circuit	
Unit III	Fast Ethernet, FDDI Token Ring, Wireless LAN, Inter-Networking Devices, Bridge, Routers Gateways, Repeater, Distance Vector Routing, Shortest Path Routing, Broadcast Routing, Multicast Routing, TCP/IP Protocol.	
Unit IV	Comparison between OSI and TCP/IP Models, TELNET, FTP, SMTP, MINE, UDP, URL (Uniform Resource Locator) HTTP differentiated services, network interconnection model, MPLS, ATM applications.	
Unit V	Network Security : Network Security Issues, Firewalls – Need and Features of Firewalls, Types of Firewall Technology- Network Level and Application Level, Cryptography, Cryptography algorithms, security protocol	

TEXT & REFERENCE BOOKS:

- 1•DATA COMMUNICATION BY PRAKASH C GPTA
- 2•COMPUTER NETWORKING BY ANDREWS TANANBAUM
- 3• UNDERSTANDING DATA COMMUNICATION OF NETWORKING BY WILLIAM A SHAY
- 4• COMMUNICATION AND NETWORK BY LEWIS MACHENZIE

BACHELOR'S OF COMPUTER APPLICATIONS

BCA 52

FIELD PROJECT/ INTERNSHIP /APPRENTICESHIP/COMMUNITY SERVICE

Note : Project work should be undertaken in an organization engaged in software/ hardware development. Synopsis of project should be approved by Principal of concerned institute. A confidential report of student related to work done by him should be taken from external project guide

1. 'Title of the Project'
2. Introduction and Objectives of the Project.
3. Relevance of the topic for the benefit of the society
4. Analysis: (DFDs at least Lip to second level, ER Diagrams/ Class Diagrams/ Database Design etc. as per the project requirements).
5. Design: A complete structure which includes: Number of modules and Purpose af each module to provide an estimation of the student's effort on the project. Data Structures as per the project requirements for all the modules:
6. Testing process to be used.
7. Reports generation (Mention tentative content of report)
8. Tools / Platform, Hardware and Software Requirement specifications
9. Are you doing this project for any Industry/Client? Mention Yes/No. If Yes, Mention the Name and Address of the Industry or Client
10. Future scope and further enhancement of the project,

BACHELOR'S OF COMPUTER APPLICATIONS

BCA 53

INFORMATION STORAGE AND MANAGEMENT

Unit I	Introduction to Storage Technology: Data proliferation, evolution of various storage technologies, Overview of storage infrastructure components, Information Lifecycle Management, Data categorization.	
Unit II	Storage Systems Architecture: Intelligent disk subsystems overview, Contrast of integrated vs. modular arrays, Component architecture of intelligent disk subsystems, Disk physical structure Components, properties, performance, and specifications	
Unit III	Introduction to Networked Storage: JBOD, DAS, NAS, SAN & CAS evolution and comparison. Applications, Elements, connectivity, standards, management, security and limitations of DAS, NAS, CAS & SAN.	
Unit IV	Hybrid Storage solutions; Virtualization: Memory, network, server, storage & appliances. Data center concepts & requirements, Backup & Disaster Recovery: Principles Managing & Monitoring: Industry management standards (SNMP, SMI-S, CIM), standard framework applications	
Unit V	Information storage on cloud :Concept of Cloud, Cloud Computing, storage on Cloud, Cloud Vocabulary, Architectural Framework, Cloud benefits, Cloud computing Evolution, Applications & services on cloud, Cloud service providers and Models	

TEXT & REFERENCE BOOKS:

1. G. Somasundaram & Alok Shrivastava (EMC Education Services) editors; Information Storage and Management: Storing, Managing, and Protecting Digital Information; Wiley India.
2. Ulf Troppens, Wolfgang Mueller-Friedt, Rainer Erkens, Rainer Wolafka, Nils Haustein; Storage Network explained : Basic and application of fiber channels, SAN, , iSES, INFINIBAND and FCOE, Wiley India.
3. John W. Rittinghouse and James F. Ransome; Cloud Computing : Implementation , Management and Security, CRC Press, Taylor Frances Pub.. Anthony T. Veleto,

BACHELOR'S OF COMPUTER APPLICATIONS

BCA 54

Software Engineering

Unit I	Introduction of system concept, characteristics of system, elements of system, The System Development Life Cycle, The Role of System Analyst. Introduction system planning & initial investigation, various information gathering tools ,feasibility study	
Unit II	Software Process, Product and Project: The Product : Software, Software Myths, The process : Software Engineering : A Layered Technology, Software Process Models, The Linear Sequential Model, The Prototyping Model, The RAD Model, Evolutionary Software Process Models	
Unit III	Software Project Planning and Design : Software Project Planning : Project planning objectives, Decomposition Techniques, Empirical estimation models, The Make/Buy Decision., Risk analysis. Software Design: Design Principles, Cohesion & Coupling.	
Unit IV	Software Quality Assurance and Testing: Software Quality Assurance : Quality Concepts, The Quality Movement, Software Quality Assurance, Software Reviews, Formal Technical Reviews, Formal Approaches to SQA, Statistical Software Quality Assurance, Software Reliability.	
Unit V	Advanced Topics: MIS & DSS: Introduction to MIS, long range planning, development and implementation of an MIS, applications of MIS in manufacturing sector and in service sector. Decision Support System concepts, types of DSS. Object Oriented Software Engineering: Object Oriented Concepts, Identifying the Elements of an Object Model, Management of Object Oriented Software Projects. CASE tools, Re-engineering	

TEXT & REFERENCE BOOKS:

1. Richard Fairley : Software Engineering Concepts (TMH)
2. Pankaj Jalote “Software Engg” Narosa Publications.
2. Ian Sommerville : Software Engineering 6/e (Addison-Wesley)
3. Richard Fairley : Software Engineering Concepts (TMH)

BACHELOR'S OF COMPUTER APPLICATIONS

A 5

Artificial Intelligence

Unit I	General Issues and Overview of AI: The AI problems, what is an AI technique, Characteristics of AI applications. Introduction to LISP programming: Syntax and numeric functions, Basic list manipulation functions, predicates and conditionals, input output and local variables, iteration and recursion, property lists and arrays.	
Unit II	Problem Solving, Search and Control Strategies: General problem solving, production systems, control strategies forward and backward chaining, exhaustive searches depth first breadth first search. Heuristic Search Techniques: Hill climbing, branch and bound technique, best first search & A* algorithm, AND / OR graphs, problem reduction	
Unit III	Knowledge Representations: First order predicate calculus, skolemization, resolution principle & unification, interface mechanisms, horn's clauses, semantic networks, frame systems and value inheritance, scripts, conceptual dependency.	
Unit IV	Natural Language processing: Parsing techniques, context free grammar, recursive transitions nets (RNT), augmented transition nets (ATN), case and logic grammars, symantic analysis .Game playing Minimax search procedure, alpha-beta cutoffs, additional refinements.	
Unit V	Probabilistic Reasoning and Uncertainty: Probability theory, bayes theorem and bayesian networks, certainty factor. Expert Systems: Introduction to expert system and application of expert systems, various expert system shells, vidwan frame work, knowledge acquisition.	

TEXT & REFERENCE BOOKS:

1. Elaine Rich and Kevin Knight “Artificial Intelligence” - Tata McGraw Hill.
2. “Artificial Intelligence” 4 ed. Pearson.
3. Dan W. Patterson “Introduction to Artificial Intelligence and Expert Systems”,
4. Nils J. Nilson “Principles of Artificial Intelligence”, Narosa Publishing House.
5. Clocksin & C.S.Melish “Programming in PROLOG”, Narosa Publishing House.

BHABHA UNIVERSITY, BHOPAL

BACHELOR'S OF COMPUTER APPLICATIONS (BCA)



SYLLABUS BCA VI SEMESTER (EFFECTIVE FROM JULY 2022)

Department of Computer Science & Applications
BHABHA UNIVERSITY, BHOPAL
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BHABHA UNIVERSITY, BHOPAL

SCHEME OF EXAMINATION BCA

Semester VI

SNO	PAPER TYPE	NO. OF PAPERS	CREDIT	LECTURE HRS PER WEEK		MAX MARKS	
				THEORY	LAB	THEORY (UNIVERSITY EXAM +CCE)	LAB UNIVERSITY EXAM +CCE)
1	MAJOR (WEB DESIGNING) BCA61	1	6	3		100 (60+40)	
	WEB DESIGNING LAB BCA65	LAB			4		50
2	FIELD PROJECT/ INTERNSHIP /APPRENTICESHIP/COMMUNITY SERVICE BCA62	LAB	6				150
3	DSE(DISCIPLINE SPECIFIC ELECTIVE) (CLOUD COMPUTING) BCA63	1	8	3		100 (60+40)	
	DSE(DISCIPLINE SPECIFIC ELECTIVE) (THEORY OF COMPUTATION) BCA64	1				100 (60+40)	
4	ADL SOFTWARE TESTING AND QUALITY ASSURANCE A6	1	4	3		100 (60+40)	
	TOTAL	4	24			400	200
	Grand Total						600

BACHELOR'S OF COMPUTER APPLICATIONS

BCA 61
WEB DESIGNING

Unit I	Internet & World Wide Web: Introduction, Internet addressing, Browsers, URL, Web Server, Web Site, Homepage, Domain Names Basic Concepts. Facilities available over the Internet – email, www, blog, chat. Components of Internet – client/server modem	
Unit II	Basic Formatting of HTML documents layout – Font, Lists, Paragraph, Break Rule, Horizontal Rule, Colors, Backgrounds, text elements, etc. Links in html anchor element and its attributes, images and anchors. HTML Media Types, Audio & Video support in web browsers.	
Unit III	Advanced Layout: Tables, Frames, Layers, Tables layout, elements and attributes. Frames using frames, layout of frames, problems using frames. Layers-concept of layers.	
Unit IV	Style sheets – Basic concept of style sheet, using style sheet, using style sheets – basics and properties, positioning with style sheets, Basic Interactivity in html: Forms- Concept of Forms, attributes, .	
Unit V	Advanced Features in HTML: Scripting, , Web publishing – Concepts. Scripting – purpose of scripting, & designing events. DHTML: Overviews of DHTML and Document object Model, html & scripting access.	

TEXT & REFERENCE BOOKS:

1. The Complete Reference to HTML – by Thomas A Powell, TMH, II Edition.
2. Using HTML – by Lee Anne Phillips, PHI
3. Learning computer Fundamentals, MS Office & Internet & web technology by Dinesh Maidasani, Firewall Media.
4. Introductions to Internet & Java – by Kiran Nayar, Sukhjeet Kaur, Kalyani Publishers.

BACHELOR'S OF COMPUTER APPLICATIONS

BCA 62

FIELD PROJECT/ INTERNSHIP /APPRENTICESHIP/COMMUNITY SERVICE

Note : Project work should be undertaken in an organization engaged in software/ hardware development. Synopsis of project should be approved by Principal of concerned institute. A confidential report of student related to work done by him should be taken from external project guide

1. 'Title of the Project'
2. Introduction and Objectives of the Project.
3. Relevance of the topic for the benefit of the society
4. Analysis: (DFDs at least up to second level, ER Diagrams/ Class Diagrams/ Database Design etc. as per the project requirements).
5. Design: A complete structure which includes: Number of modules and Purpose of each module to provide an estimation of the student's effort on the project. Data Structures as per the project requirements for all the modules:
6. Testing process to be used.
7. Reports generation (Mention tentative content of report)
8. Tools / Platform, Hardware and Software Requirement specifications
9. Are you doing this project for any Industry/Client? Mention Yes/No. If Yes, Mention the Name and Address of the Industry or Client
10. Future scope and further enhancement of the project,

BACHELOR'S OF COMPUTER APPLICATIONS

BCA 63
CLOUD COMPUTING

Unit I	Cloud Computing Fundamentals: Cloud Computing definition, Types of cloud, Cloud services, Benefits and challenges of cloud , Evolution of Cloud Computing, architecture of cloud computing, Applications cloud computing.	
Unit II	Types of Computing and Clouds: Cluster Computing, Grid Computing, Grid Computing Versus Cloud Computing, Key Characteristics of Cloud Computing, Cloud Models, Benefits of Cloud Models.	
Unit III	Cloud Services and File System: Types of Cloud services: Software as a Service - Platform as a Service – Infrastructure as a Service - Database as a Service- Monitoring as a Service – Communication as services and Service providers.	
Unit IV	Virtualization: Basics of Virtualization, Types of Virtualization, Implementation Levels of Virtualization, Virtualization Structures, Tools and Mechanisms ,Memory, I/O Devices and OS.	
Unit V	Security in the Cloud: Security Overview – Cloud Security Challenges and Risks – Software-as- a-Service Security – Security Monitoring – Security Architecture	

TEXT & REFERENCE BOOKS:

1. Cloud Computing "A Practical Approach" Anthony T. Velte, Toby J. Velte, Robert Elsenpeter. McGraw Hill. Kai Hwang, Geoffrey C Fox, Jack G Dongarra,
2. "Distributed and Cloud Computing, From Parallel Processing to the Internet of Things", Morgan Kaufmann Publishers, 2012.
3. John W.Rittinghouse and James F.Ransome, "Cloud Computing: Implementation, Management, and Security", CRC Press, 2010.
4. Ronald L. Krutz, Russell Dean Vines, "Cloud Security – A comprehensive Guide to Secure Cloud Computing", Wiley – India,

BACHELOR'S OF COMPUTER APPLICATIONS

BCA 64
THEORY OF COMPUTATION

Unit I	Review of Mathematical Preliminaries : Set, Relations and functions, Graphs and trees, string, alphabets and languages. Principle of induction.	
Unit II	Theory of Automation : Definition, description, DFA- Deterministic Finite Automata ,NFA- Non-Deterministic Finite Automata, Transition systems,2DFA, equivalence of DFA & NDFA.	
Unit III	Regular expressions, regular grammar, FSM with output (mealy and moore models), Minimization of finite automata. Formal Languages : Definition & description. finite automata, equivalence of FA and regular expression, equivalence of two way finite automata, equivalence of regular expressions.	
Unit IV	Turing Machine : Model, design, representation of TM, language accepted by TM, determine & non-determinism in TM, Two way infinite tape, Halting problems of TM	
Unit V	Computability : Concepts, Introduction to complexity theory, Introduction to undecidability, recursively enumerable sets, primitive recursive functions.	

TEXT & REFERENCE BOOKS:

BOOKS

1. Hopcroft & Ullman "Introduction to Automata theory, languages & Computation" , Narosha Publishing house.
2. Lewish Papadimutrau "Theory of Computation" , Prentice Hall of India, New Delhi.
3. Peter linz, "An Introduction to formal language and automata", Third edition, Narosa publication

BACHELOR'S OF COMPUTER APPLICATIONS

A 6

SOFTWARE TESTING AND QUALITY ASSURANCE

Unit I	Software Testing Process Objectives Testing Techniques Software Testing Life Cycle Concept of Testing Types of Errors Stubs and Drivers Verification and Validation Different Types of Verification & Validations Mechanisms Concepts of Software Reviews Code Inspection and Code Walkthrough Testing of Component Based Software System Energy Efficient Testing Mobile Application Testing.	
Unit II	Software Testing Methods Testing Fundamentals Test Case Design White Box-Testing and its Types Black Box Testing and its Types Software Testing Strategies Strategic Approach to Software Testing UNIT Testing Integration Testing Validation Testing System Testing Test Planning Budgeting and Scheduling	
Unit III	Software Testing Metrics Concept and Developing Testing Metrics Different Types of Metrics Complexity Metrics Defect Management Definition of Defects Defect Management Process Defect Reporting Metrics Related to Defects Using Defects for Process Improvement.	
Unit IV	Software Quality Factors Affecting Software Quality Models Software Quality Estimation Quality Metrics Quality Assurance SQA Activities Software Reviews Formal Technical Reviews Quality Control Quality Management and SQA Plan. Quality Improvement Pareto Diagrams Cause-Effect Diagrams Scatter Diagrams Run Charts Total Quality Management Statistical Quality Assurance Software Reliability the ISO 9001 Quality Standard Six Sigma Informal Reviews.	
Unit V	Quality Costs Quality Cost Measurement Utilizing Quality Costs for Decision-Making. Manual Vs Automatic Testing Basics of Automated Testing Drawback of Manual Testing Advantages of Automation of Testing Factors for Automation Testing Types Automation of Testing Tools Introduction to QTP. IDE Basic Components in QTP. QTP Framework Write Scripts. Introduction to Software Testing Tools.	

TEXT & REFERENCE BOOKS:

1. Roger S. Pressman "Software Engineering-A Practitioners Approach" McGraw Hill
2. K.K. Aggarwal & Yogesh Singh "Software Engineering 2nd Ed. New Age International
3. Kshirsagar Naik PriyadarshiTripathy "Software Testing and Quality Assurance" Theory and Practice Wiley-Spektrum. I Edition August 18 2008.